



Smart Agriculture Architecture and Current State in India

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Abstract--The IoT (Internet of Things) is going hand in hand with every domain. The agriculture domain is also set to exploit the benefits of IoT. Smart agriculture, Precision farming are the buzzwords in the agriculture domain. It will help farmers in getting more and more benefits of their crops and can also help them in avoiding many issues related to weather and soil conditions which are harmful to their crops. Farmers can keep an eye on various factors like changing weather conditions, soil and fertilizers, pest control, cattle, etc. by using the precision farming concept. Use of smart sensors and analyzing data received from them will lead to the precision farming concept. Precision farming is the latest concept in the agriculture field whereby making use of wireless technologies one can do precise and controlled farming. Agriculture is an important pillar of the Indian economy hence integrating technology with agriculture will give rise to a new era and will strengthen the economy as well. This paper is outlined to discuss the role of IoT in the Agriculture domain, Smart Agriculture Framework, and the analytics that can be done on data accessed through smart devices. The paper will also outline current state of smart agriculture in India.

Keywords:

IoT, Agriculture Domain, Precision Farming, Smart Agriculture, Internet of Things

I. INTRODUCTION

The IoT (Internet of Things) is a set of interconnected devices which are basically mechanical or digital in nature. All such devices have their unique identification number. These IoT devices are capable enough to transfer information or data over the network and do not require any human intervention for the same. IoT is being used in versatile domains like smart homes, smart classes, health care, business, and agriculture. The traditional IoT architecture collects data from sensors, actuators, wearables, and others dispersed IoT devices. The collected data is being transmitted to the central repository, where it is processed collectively or transmitted further.

IoT is changing the way each domain is working now a days. The agriculture industry also needs to draw benefits of IoT and should make remarkable success by overcoming various challenges of weather changes and other environmental factors. Incorporating IoT will lead to the smart farming concept and it will enable the agriculture industry to increase

operational efficiency, lower costs, reduce waste, and improve the quality of their yield.

In today's scenario, with the increasing population, more food is required, and to fulfill the demand the agriculture industry and farmers need to work in smartly. Despite the various weather conditions, rising temperature, and weather fluctuations, farmers need to adapt the latest technology to face these challenges. Using this technology productivity of the farmers may be improved.

The smart farming concept is ready to change the entire concept of traditional farming. The smart farming system works in a way where every activity related to crops and fields can be monitored. The sensors can record light, humidity, temperature, soil moisture, etc., and help farmers to monitor their crops from anywhere and take appropriate steps to improve their crop performance. IoT is not going to help farmers to improve traditional farming ways only. The new concepts also originate with the usage of these sensors and actuators like organic farming, family farming preservation of high-quality varieties etc. Smart farming provides great benefits like water usage in an efficient manner or optimization of various inputs.

IoT is being used in agriculture in many ways, and it converts normal farming to smart farming. There are many different techniques or technology by which IoT comes into the picture and plays a vital role in the Agriculture domain. It makes farming more precise and hence known as Precision Farming.

II. LITERATURE REVIEW

Various researchers have worked in this area and proposed algorithms and approaches to meet smart farming demands. Liu et al. [1] suggested approach which is based on precision irrigation. Ersin et al [2] proposed an irrigation system that makes use of a microcontroller. They justified their proposed system is quite efficient as compared to other conventional methods. Agarwal et. al.[3] proposed smart irrigation system. They utilized Raspberry pi and Arduino for developing such kind of smart system. Koprda et al.[4] also proposed a similar kind of smart irrigation system based on the microcontroller concept. Ahouandjinou et al. [5]



demonstrated the usage of farm pest detection by making use of ultrasonic sensors. Goap et al. [6] proposed IoT based agriculture system, whereas Smith et al. [7] used Machine Learning approaches for soil classification. Kwok et al. [8] suggested plant detection using deep learning and then, based on plant type, its appropriate irrigation amount required. Deep learning and altitude-based economic irrigation approach has been discussed in the research by Wang, Muzzammel, Raheel et al. [9]. The study of current literature and work of various researchers emphasized the active role of IoT and Machine Learning techniques in Smart Farming.

III. MAJOR APPLICATIONS OF IoT IN AGRICULTURE

IoT can be well exploited by using of upcoming techniques like Machine Learning, Deep Learning, Data Analytics to improve the performance of the Agriculture domain and give better results to the farmer. The latest wireless IoT sensors can be integrated with the agriculture setup in the current scenario to increase agriculture performance.

A. Precision Farming

It is one of the most popular applications of IoT in agriculture. Precision farming, as in literal terms, refers to accurate and controlled farming. It uses sensors, control systems, robotics, and autonomous vehicles to perform a variety of activities. By using high-speed internet, mobile devices and satellites that are low in cost can lead to precision farming concept.

B. Livestock Monitoring

The farmers who have large farm setups may use IoT setup to track the data regarding the location and health of their cattle. The information collected through the sensors help them to visualize and identify the animals who are not well. They can further keep sick animals separate from the healthier one and thus stop spreading the disease.

C. Field Observation

By making use of IoT sensors, farmers can keep track of updated information about their fields. Complete information about their fields can be stored at one central location and easily accessed by farmers to make quick decisions accordingly.

D. Climate Conditions

Agriculture is mainly dependent upon climatic conditions. Smart sensors can be implemented to mount weather stations to get updated and the latest information regarding the weather. These sensors are placed across the field and collect data from a field which is further transmitted to the

cloud. The further analysis of such data will help in taking decisions for specific type of crops and to decisions to improve the quality and quantity of crops. Few such IoT devices are all METEO, Smart elements and Pycno.

E. Smart Greenhouses

Smart greenhouses can be implemented with the help of IoT sensors. These smart greenhouses are capable in managing climatic conditions without human intervention. Environmental conditions in a smart greenhouse can be managed very easily from the remote location as well. These sensors eliminate the need of manual monitoring. Cloud servers can be used for processing of data and to also apply control action. Such effective designs give good solutions to farmers without having manual intervention. There are companies which are providing solar powered IoT sensors that can be used to monitor water consumption by messages to farmer through SMS alerts.

IV. SMART AGRICULTURE AND DATA ANALYTICS

A. Precision Farming

Soil features analysis is the most important aspect of agriculture. Soil features analysis gives direction to farmers to take different decisions regarding their crops. Soil nutrient level decides the fate of the crops. Features like a type of soil, application of fertilizer, cropping history, level of irrigation, topography, etc., give deep insight about the various factors related to soil. In the current situation many manufacturers are providing smart sensors that farmers can use to keep a track of various soil features and take further decisions based on its status. Agrocare is one of the solutions for farmers. They can analyze multiple soil samples and make further decisions [10]. Drought is a crucial factor that affects agriculture greatly. To deal with this issue, Authors [11] used SMOS L2 to calculate the Soil Water Deficit Index (SWDI) in Spain. In [12], used MODIS (Moderate Resolution Imaging Spectroradiometer) to record various soil properties to estimate the risk of land degradation. [13] used sensor that is vision-based robot known as Agribot for sowing seeds.

B. Irrigation Data

The current scenario of the water crisis puts an utmost demand for its proper utilization in the agriculture field. The required amount of water should be provided to the crops at the required time. Estimation of accurate water demand of a crop is a difficult task as many other factors are also associated with it. A precise wireless system that can provide accurate water demand will lead to a better crop health. A significant increase in crop efficiency is expected with the use of IoT-based techniques. One example for such a case is irrigation management by assessing crop water stress index [14]. To calculate CWSI, one needs to record the crop canopy of different time zones and temperatures. Intelligent farm monitoring systems use field sensors connected to central server meant for processing and analysing data collected from farm sensors. A prominent example is VRI (Variable Rate Irrigation) optimization by CropMetrics [15],



that performs based on variability of soil and helps in efficiency of water usage.

C. Fertilizer

A balanced quantity of nutrients is required for healthy crops. Improper quantity of nutrients will spoil the quality of crops and soil also many times. In smart agriculture, the exact amount of nutrient requirements can be traced, reducing the risk of negative effects on soil and crop. Latest IoT sensors based approaches help in estimating the nutrient requirements of the crops with better accuracy.[16,17] One example for that is NDVI (Normalized Difference Vegetation Index), which uses of satellite images to monitor nutrient requirements of crops[18,19]. Besides these other benefits of IoT are fertigation [20] and chemigation; these techniques used fertilizers, pesticides, etc., through irrigation system only.

D. Forecasting of Crop Yield

Monitoring the crop yield is the most crucial aspect of Smart Agriculture/ Precision farming. Analysis of crop yield, quality of grains and moisture content directs the farmers to make appropriate decisions. Yield monitoring helps farmers in better planning about their crops. The yield quality depends on various parameters. One of them is the quality of pollens that play an important role in predicting of seed yields under changing environment. [21-23]. Analysis of quality of and maturity of yield helps farmers determine the correct time of harvesting as well. FarmRTX App can be linked to a yield monitor developed by [24] that can be installed for yield monitoring. It provides the live status of harvest.

V. SMART AGRICULTURE ARCHITECTURE

Smart agriculture architecture may have the following layers-

- Innermost layer (Sensor Layer)
- Middle layer (Network Layer and Service Layer)
- Topmost layer (Application Layer)

Sensor Layer is the most important layer of IoT-based smart farming architecture. There are various sensors that can be placed in smart farming architecture and can trap variety of data. These sensors are meant to keep track of moisture, soil chemical properties, pH level, and many more important data. In smart farming architecture variety of sensors are placed to measure different crop attributes.

Optical sensors- are used to record various properties of soil. These sensors used light reflection information to capture information that data is further analyzed with photogrammetry techniques. The information may be related to moisture, color of soil, vegetation, temperature, height, size, and position of plants.

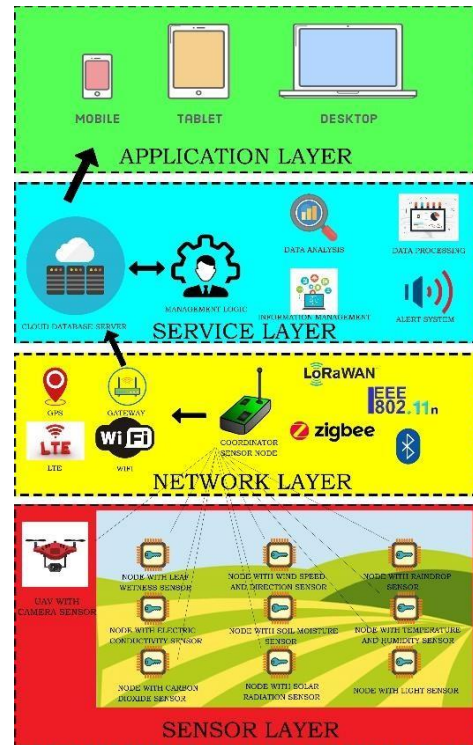


Fig. 1 Smart Agriculture Architecture

Electrochemical sensors are used to collect data regarding soil nutrients and their pH level. Location sensors can be used to access spatial information using GPS satellites. Weather stations are different units which are placed at a different location in the farming area. These stations trap data related to airflow, wind speed, humidity level, solar radiation, direction of a wind, etc.

Network Layer- WSN (Wireless Sensor Network) communication protocols are used at this layer. WSNs facilitate monitoring of crop quality as well as surveillance of crops. WSN helps in regular monitoring of various environmental factors and helps the farmer in achieving maximum productivity. The sensors at this layer used different topologies depending upon application like precision farming, field irrigation management, or crop management. For smart agriculture, there is no specific communication technology and routing protocol. There is a vast variation in usage of various communication technologies. There are a variety of protocols that work in precision agriculture. Few of them are like- ZigBee, LoRa, 6LoWPAN, RFID etc.

Service Layer makes an important component of smart agriculture architecture. The processes at the service layer enable the advanced cultivation procedure, and it also helps boost productivity. Information Management is deployed at the service layer. The farmers can make use of the data stored at service layer. Farmer can collect data from various WSN nodes and can analyze them as well by using historical data as well. Mahout and other IoT platforms help in performing BigData Analytics in the agriculture domain. The data mining process is



based on Hadoop or Apache Spark, which can find hidden patterns in the collected data.

Application Layer provides the visualization results after the review and data analytics performed by the service layer. The application layer is responsible for presenting the results in a user-understandable way, and it will help in field optimization, pesticide drift control, to name a few. There are a variety of applications that are used in smart agriculture framework at application layer like- Variable Rate Fertilizer, Variable Spraying application, Weed Mapping Application, and other IoT based applications help farmers in decision making.

VI. CHALLENGES FOR SMART FARMING

Smart farming is a new vibe that utilizes the latest technology trends like AI, IoT, Machine learning to improve the agriculture performance. The important aspect of smart farming is implementing various sensors and other devices used to collect relevant data required for analysis purposes. Technology is changing very fast; hence to cater to the latest updations in the architecture is a difficult task. Besides that, network bandwidth is also one of the issues in the implementation of smart farming. There are various other challenges or issues that smart farming faces-

Standards interoperability

Interoperability is the major concern while implementation of Smart Farming architecture. Technology is growing leap and bounds. Every day, some new IoT tools and platforms are hitting the market. These tools and platforms do not follow the same standards, resulting in interoperability issues that lead to lack of uniform data analysis and visualization.

Lack of sufficient knowledge

At the center of smart farming we have farmers who are not very comfortable with these technological aspects. Training farmers for the basic concepts and utilization of various smart devices is of utmost importance. Lack of knowledge will lead to disastrous results.

Internet Connectivity

A major hurdle in the implementation of smart farming architecture is the poor internet connectivity in rural areas. Poor network bandwidth issues also make the implementation of smart farming architecture quite tricky.

Scalability

Farmers have a varied size of farms; hence they need IoT devices that should be scalable, which means that the same technology should apply to all sizes of farms. These devices are, in general, configured manually; hence it is also a concern area. The devices and tools used for smart farming should be self-configurable to draw maximum benefits.

Energy Consumption

It is another critical challenge in the case of smart farming. To implement smart farming architecture, many data centers, gateways, and hubs are required to make sensors operable and lead to heavy energy consumption. It needs more energy sources to restock that energy.

Technology failure

Too much dependency on technology will lead to some serious risks for the crops too. Failure of technology at any point in time will lead to crop damages and can affect farmers. For example, failure of smart sensors can lead to overwatering or under watering of crops and can have serious impact on crops.

Security Concerns

Security is a significant concern and challenge as well in smart farming. Data theft, As the count of middleware technology, endpoints, and IoT devices in active use in agriculture is increasing, the number of entry-points for malicious third-party programs is going up as well. Since the third-party attacks on a complex IoT system are often decentralized, detecting, and removing them emerges as a big challenge.

VII. CURRENT STATE OF SMART AGRICULTURE IN INDIA

Precision farming is a base of Smart Agriculture. Precision farming is dependent upon GPS (Global Positioning System) technology which makes use of sensors. Precision agriculture provides accurate information about the requirements of water, fertilizers, pesticides, etc. The grain requirement of India is going to reach 480 million tons by the year 2050. To attain such requirement, there is an utmost need to adapt the concept of Precision farming. Applying various precision farming technologies in a balanced way will lead to promising results in India as well. Precision farming can be categorized into soft and hard. Soft precision farming involves human intervention for management and decision-making. However, hard precision farming is all dependent upon GPS, RS (Remote Sensing), VRT (Variable Rate Technology), etc.

Being the second largest producer of sugarcane, India should adapt precision farming concept for it and harvest the benefits of it. Besides that, in other fruit and vegetable farming has already started adapting these automated systems for better production results.

A. Precision Farming Adoption in India

India should start adapting the precision farming concept for small farms where sensors and small robots will contribute. These may run on various renewable fuels like



biofuels and also on electricity that may be produced by agriculture residues. The US and Europe had already adapted small robots to get precise irrigation water requirement, weed removal and harvesting. India should also look forward in this direction and should adopt such techniques. Japan and the US utilized drones for disease identification in farms. India is the ideal country where small farms can introduce these concepts of precision farming.

Though there are high precision farming profits, but still it is in a nascent stage in India. Many companies like TKK (Tata Kisan Kendra), Trimble and Fasal that are working on this concept and will soon be introduced to the farmers in India. The TCL (Tata Chemicals Limited) is already working in this direction and trying to convert the whole farming business by using of advanced techniques. TCL is already propagating its services through TKK to use and understand remote-sensing techniques for assessing of soil condition, health of crop, prediction of crop yield and pest invasions.

VIII. CONCLUSION

IoT is having promising role in agriculture domain. Various researchers had justified its importance in agriculture domain. Agriculture plays an important role in economy of various countries. Performance improvement in farming by making use of IoT is setting a new benchmark. Latest discoveries of sensors and upgraded protocols are now set to support agriculture domain. Agriculture industry should exploit the benefits of these latest technologies and should improve the performance level.

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